

CHAPTER 16

Russia Today



GeoJournal

As you read this chapter, make notes in your journal about life in Russia today. Use clear, specific language to explain how recent economic and political changes have affected the people of Russia.



Chapter Overview Visit the **Glencoe World Geography** Web site at tx.geography.glencoe.com and click on Chapter Overviews—Chapter 16 to preview information about the region today.

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CONTENTS

Guide to Reading

Consider What You Know

Russia continues to adjust to dramatic and sometimes difficult political and economic changes. How do you think these changes probably affect Russians' attitudes toward the old Soviet government?

Read to Find Out

- How has Russia made the transition to a market economy?
- How have agriculture, industry, transportation, and communications in Russia changed since the breakup of the Soviet Union?
- What is Russia's relationship to the global community?

Terms to Know

- command economy
- consumer goods
- black market
- market economy
- privatization
- kolkhoz
- sovkhoz

Places to Locate

- Siberia
- Vladivostok

◀ *Teens walking through Red Square, Moscow*

Living in Russia



A Geographic View

The Price of Freedom

Not long ago . . . I met a woman named Larissa Pavlova. She was a teacher who now sold old clothes evenings and weekends to supplement her family's income. Countless thousands of Muscovites work second and third jobs to get by. . . . "Moscow is filled with what our good Comrade Lenin called contradictions," she said. "The rich get richer and the rest of us tread water or drown. I work much harder than I did in the old days, and sometimes that makes it hard to remember what we've gained. Freedom is sweet, but it's also a heavy, heavy load."

—David Remnick, "Moscow: The New Revolution," National Geographic, April 1997



Muscovites selling food on the street

Russians hoped that the end of the Soviet-controlled economy and the birth of Russian independence would bring quick and painless economic change. As the teacher from Moscow discovered, however, shifting toward a freer economy could also bring hard times. Russia continues its efforts to create a working economy that will provide for its people and maintain its place in the global marketplace.

Changing Economies

Since the fall of communism, Russia has faced many economic challenges, such as providing more jobs for its citizens, increasing food production at home, and expanding trade internationally. As Russia works to strengthen its economy, its citizens also face ethnic unrest, rising crime, and declining health and social services.

The Soviet Command Economy

Under Communist leaders, the Soviet Union operated as a command economy. In a **command economy**, a central authority makes key economic decisions. The government owned banks, factories, farms, mines, and transportation systems. Members of the State Planning Committee, known as *Gosplan*, decided what and how much to produce, how to produce it, and who would benefit from the profits. *Gosplan* also controlled the pricing of most goods and decided where they would be sold.

The Soviet government emphasized heavy industry—the manufacture of goods such as tanks and other military hardware, machinery, and electric generators. As a result, the Soviet Union became an industrial giant and a world power, but its people could not buy many **consumer goods**, or goods needed for everyday life.

Unemployment in the Soviet Union was low, but so were wages, because most men and women worked at state-run factories and farms. People often could not afford the few consumer goods that factories produced. Even when people had enough money, such goods were hard to find. Some items could be bought on the **black market**, an illegal trade

in which scarce or illegal goods are sold at prices even higher than those set by the government. Most workers, however, could not afford to pay such high prices with their limited incomes.

By the 1970s and 1980s, Western countries and some Asian countries had turned away from heavy industry to focus on computer technology and global communications. The Soviet system during this time, however, focused on increased industrial production and did not invest in developing new high-technology industries. As a result, the Soviet Union's economy stagnated, and its standard of living declined while the global economy entered a dynamic new era of change.

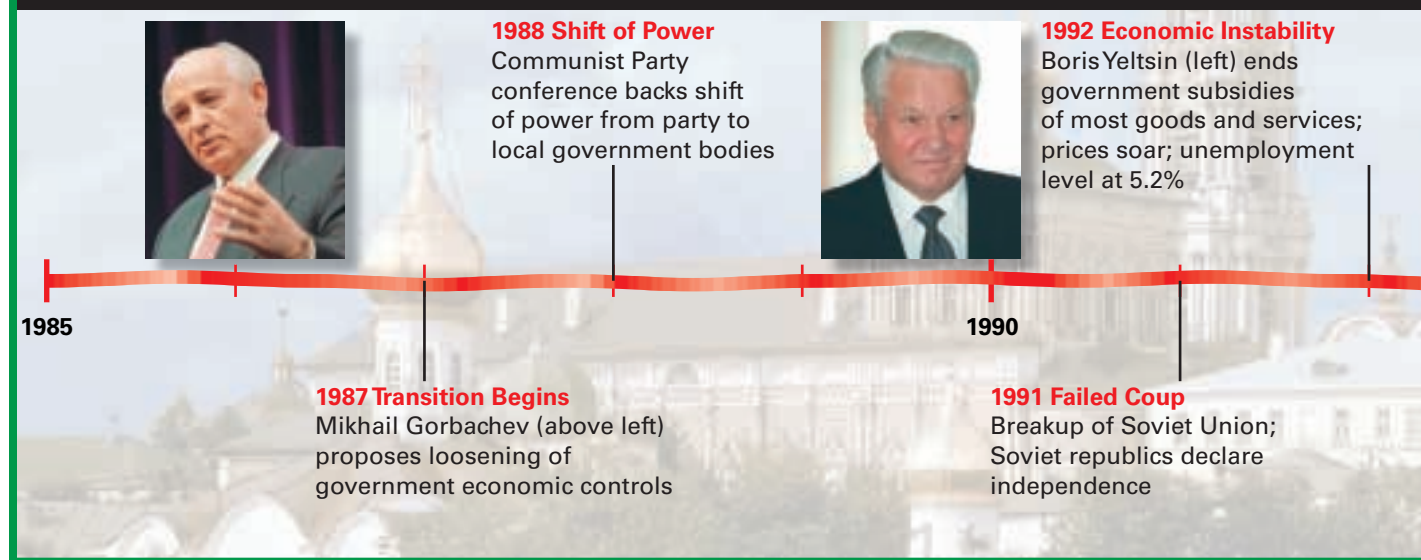
The Market Economy

When Mikhail Gorbachev came to power in 1985, the Soviet Union's economy was in serious trouble. To remedy the crisis, Gorbachev began to move away from a command economy toward a **market economy**, in which businesses are privately owned. Production and prices in a market economy depend on supply and demand. Things offered for sale are *supply*; people's desire to buy those things is *demand*. As part of his program of



CHART STUDY

Russia's Road to a Market Economy



perestroika, or restructuring, Gorbachev reduced some government controls, allowed people to start small businesses, and encouraged foreign investment. Boris Yeltsin, Gorbachev's successor, expanded this process.

Economics

Privatization

Russia's economy continued to change after the Soviet Union officially ceased to exist in 1991. When Russia and the other Soviet republics became independent, they eliminated most remaining economic controls. Russian President Boris Yeltsin removed 90 percent of price controls and encouraged the mass **privatization**—a change to private ownership—of state-owned companies and industries, such as mining and oil extraction and processing. This process of privatization favored important businesspeople, political insiders, and foreign investors, all of whom could afford to purchase large companies. Rather than reinvest in Russia and its economy, many of these people invested their profits outside the country. Most average Russian workers did not benefit from this changing economic system: they neither earned nor were spending the new wealth.

“By 1995 privatization had gained a negative reputation with ordinary Russians, who coined a slang word *prikhvatizatsiya*, a combination of the Russian word for ‘grab’ and the Russianized English word ‘privatize,’ producing the equivalent of ‘grabification.’”

Glenn E. Curtis (ed.), *Russia: A Country Study*, 1996

Widespread corruption complicated the privatization process in the new Russia. Organized crime groups and corrupt public officials operated throughout the country, especially in Moscow. Some people grew rich through special government favors that allowed them to buy property at far below its true value. This illegal behavior damaged the economy and absorbed investment funds that could have been used to rebuild the country.

The Transition Continues

The Russian economy experienced ups and downs throughout the 1990s. Russians could find

1994 Unemployment Rises

Unemployment level rises to 7.7% amidst corruption and a struggling economy



1998 Financial Crisis

Ruble plummets; Russia cannot make debt payments; unemployment reaches 12.3%

2000 Putin Takes Control

Cracks down on corruption; announces program of tax cuts, private land ownership, and greater public access to banks

1995

2000

1993 Privatization

Drive is launched to privatize small- and medium-sized businesses

1996 Unemployment Rises

Unemployment climbs to 9.9%

1999 Economic Recovery

Russian economy begins recovery; unemployment levels off at 12.4%

Geography Skills for Life

1. **Interpreting Charts** When did the transition to a market economy begin?
2. **Applying Geography Skills** How did political change affect economic development and the standard of living in Russia during the 1990s?



Geography Skills for Life

Kaliningrad, Russia To supplement their income, many Russians sell hard-to-find goods, such as car parts, in open-air markets.

Place Describe the characteristics of Russia's economy in 1991 and today.

more consumer goods in shops. However, without controls, prices soared, and many people could not afford to buy the goods that were available. Between 1990 and 1995, the total value of goods and services produced in Russia fell by 50 percent, a far greater drop than the United States experienced during the Great Depression of the 1930s. Following a 1998 financial crisis, the ruble, Russia's currency, lost 71 percent of its value. Prices, which had dropped, rose once again. The international community made large loans to aid the Russian economy.

Yeltsin resigned as president of Russia in 1999. His successor, Vladimir Putin, inherited an unstable economy. Russia's economy needed money and a stronger banking system, which would help keep more Russian money at home. Putin also needed to improve the Russian military. However, he needed to do so without overspending on the armed forces at the expense of overall economic growth, a problem experienced previously by Soviet leaders.

An inefficiently managed government, budget shortfalls, unclear property rights, an unstable currency, corruption, and organized crime all threaten Russia's economic stability. However, there is also potential for success. Russia can rebuild its economy by relying on its vast natural resources, developed industries, and well-educated citizens.

Agriculture and Industry

Under the Soviet system, farms were organized into state-controlled kolkhozes (kahl•KAW•zehz) and sovkhozes (sahf•KAW•zehz). The **kolkhozes** were small farms worked by farmers who shared, to a degree, in the farm's production and profits. **Sovkhozes** were large farms run more like factories, with the farm workers receiving wages. However, prices and production in both the agricultural and industrial sectors were controlled by the government. Both the agricultural and industrial sectors suffered because the system did not motivate workers. As a result, long before the 1980s Soviet agriculture did not produce enough food to feed its people, and the government had to import additional grain and other foods.

In 1991 President Yeltsin started to restructure state-run farms so they could function better in a market economy. However, Russian farmers—accustomed to the stability of Soviet controls—continued to operate many of Russia's farms as kolkhozes or sovkhozes. Most farmers could not afford to buy land, and they worried that wealthy Russians or foreign investors might use the land for nonagricultural development. Because of these concerns, progress toward a market economy for agriculture has been slow, and crop and livestock production has fallen. Recently, however, gains in farm productivity have helped reduce the need for agricultural imports.

Like agriculture, Russian industry has also been transformed since the early 1990s. For many years



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Russia's state-owned aerospace industry and its military-industrial system were its economic and technical focus. Many of these components have become privately owned and provide export income. Russia has also encouraged foreign investment by selling shares of ownership in some Russian companies and by opening Russia's markets to Western companies. A popular American fast-food chain, for example, now has 52 restaurants in 17 Russian cities.

Russia's most important industry is petroleum extraction and processing, and the country is one of the world's largest producers of crude oil. Russia's domestic oil provides its other industries with vital energy at a reasonable cost. The country is also a major producer of iron ore, manganese, and nickel. Huge forests in Russia produce one-fifth of the world's softwood, and Russian fish-factory ships process catches from both the Atlantic and Pacific Oceans. Other major manufacturing industries include steel milling; auto and truck production; aircraft construction; and the manufacturing of chemicals, heavy machinery, and agricultural equipment. Most of these industries are in the Volga Valley,

near Moscow and St. Petersburg, and in the Ural Mountains. Although Russia's industries still face difficulties, industrial production is now rising.

Transportation and Communications

Russian transportation and communications systems lag behind those of most of the world's developed countries. In an age of speedy transportation, the Internet, and a global economy, Russia struggles to find funds for new highways and high-tech communications.

Transporting Goods

Russia's transportation systems must move food and other resources great distances to reach consumers. A major highway system links Moscow with other major Russian cities, but many roads are in poor repair. Harsh winters in places like **Siberia** often make roads impassable.

Because of its great size and climate extremes, Russia depends on railroads and waterways for most of its transportation needs. Not surprisingly,

Russia boasts the world's longest continuous railroad line. The Trans-Siberian Railroad is the greater part of the rail route from Moscow through the Siberian steppes to the Pacific port city of **Vladivostok**. Major cities are found where the Trans-Siberian Railroad crosses large rivers, such as the Ural, Irtysh, Ob, and Yenisey. Millions of tons of materials travel along thousands of miles of navigable inland waterways, which connect seaports and inland cities.

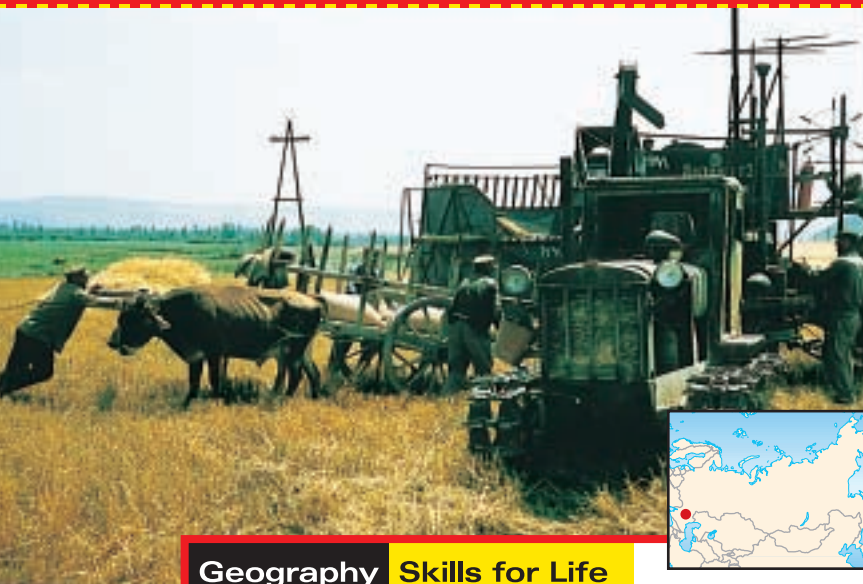
Transporting People

Most Russians live in cities and many do not own cars, so public transportation, such as trains, buses, and, in several large cities, subways, is common. Private car ownership doubled in the 1990s, but public transportation remains a practical option for Russians, in part because the government helps pay for it. The systems and equipment, however, need repair and improvements.

The Soviet Union used jet airplanes for passenger traffic, and the government



World Explorer



Geography Skills for Life

Russian Agriculture Outdated farm equipment makes farming labor-intensive for humans and animals.

Place Why have farmers in Russia been reluctant to accept a market economy system?



Geography Skills for Life

Russian Waterways Cargo cranes along the Pregolya River in Kaliningrad assist in shipping goods for export.

Place What goods does Russia export?

financially supported air travel for many years. The passenger airline Aeroflot was once the only one in the Soviet Union and at its peak carried 135 million people a year. After the fall of the Soviet Union, Aeroflot split into numerous smaller airlines. The high cost of fuel and reduced passenger traffic because of skyrocketing ticket prices have forced about 60 percent of Russia's airports to close.

Transporting Energy

Russia's large size also affects how it transports natural gas, crude oil, and other petroleum products. Pipelines are effective, although constructing and maintaining them can be difficult in areas of harsh climate. A complex maze of pipelines crisscrosses Russia, providing major Russian cities and parts of western Europe with fuel.

The oil pipelines run through Chechnya and Dagestan (DAH•guh•STAHN), ethnic republics in

southwestern Russia. Because people in these republics are fighting for their independence from Russia, control of the area's oil reserves and working pipelines is a major concern.

History

Mass Communications

Under the Soviet Union, the state owned and controlled all the mass communications systems, including newspapers, magazines, television, the postal service, and the telegraph and telephone systems. State agencies reviewed all print and broadcast materials to make sure that they contained no criticism of the government. Since the breakup of the Soviet Union, Russians have heard and read new voices and fresh views. Most families own radios and television sets, and by 1995 Russians could choose from among 10,000 newspapers and journals.

Telephone service in Russia has also grown. As a result of Russia's vast size, only 22 percent of rural households have telephones, compared with 56 percent of urban households. However, communications companies are increasingly offering services such as the Internet, e-mail, and cellular phones. These advances in communications systems will make

vital contributions to the successful transition to a market economy.

Global Interdependence

After independence Russia and the other former Soviet republics began to increase their interdependence with other countries. By expanding international trade and building political and financial relations, Russia has increasingly focused on becoming a full partner in the global community.

Trade

Russia has already established trade relations in world markets and is a major source of energy and fuels, which make up 48 percent of its exports. Lumber, metals, and chemicals are also important Russian exports. The United States, the European Union, the other former Soviet republics, China, and Japan are among Russia's major trading partners. These countries provide Russia with the consumer goods, medicines, meat, and sugar it needs.

Energy is expected to remain Russia's main item of international trade until its manufactured goods, such as machinery and light industrial products, improve in quality and become more competitively priced. Working to strengthen its industries, Russia became a member of the Asia-Pacific Economic Cooperation (APEC) forum in 1998. Negotiations

are continuing for Russia's membership in the World Trade Organization (WTO). As Russian manufacturing makes further gains, these trading networks will become even more important for the Russian economy.

International Relations

Despite its political and economic challenges at home, Russia maintains its important role in world affairs. Russia benefits from occupying the former Soviet Union's seat in the United Nations Security Council. The country has also joined European organizations that support security and cooperation. Russia has helped settle conflicts and has supported peace efforts in several countries, especially in former Soviet republics. Even as Russia asserts itself internationally, however, economic problems have drained money from its military. As a result, military forces have old equipment, and soldiers' morale is low.

Adequate financial resources are vitally important to Russia's stability and progress in the global community. Other countries and world organizations have provided loans, and foreign investors have made funds available to Russian industry. With foreign help, Russia is trying to create secure and workable systems for banking, farming, manufacturing, transportation, and communications. Although Russia has a long way to go, the economic gains made in recent years are positive signs.

TAKS Practice

SECTION 1 ASSESSMENT

Checking for Understanding

- 1. Define** command economy, consumer goods, black market, market economy, privatization, kolkhoz, sovkhoz.
- 2. Main Ideas** On a chart like the one below, fill in details about agriculture, industry, transportation, and communications in the Soviet command economy and the Russian market economy.

	Soviet Command Economy	Russian Market Economy
Agriculture		

Critical Thinking

- 3. Predicting Consequences** How might Russia's agricultural and industrial sectors be affected by Russia's growing global interdependence?
- 4. Comparing and Contrasting** How did the Soviet command economy and the Russian market economy affect the Russian people?
- 5. Making Inferences** What can you infer about Russia's goals, based on changes in Russia's trade and international relations since the Soviet breakup?

Analyzing Maps

- 6. Human-Environment Interaction** Study the economic activity map on page 341. In what area is the raising of livestock concentrated? How is this related to the physical geography of the region?

Applying Geography

- 7. Effects of Size and Distance** Think about the physical geography of Russia. Write a paragraph analyzing how Russia's vast size affects the availability of natural resources and the country's ability to develop them.



Russia's Supertrawlers:

More than a million fishing vessels scour the oceans for fish. As a result, fish populations are shrinking worldwide. Enormous ships called supertrawlers are largely to blame. Towing huge trawl nets—some large enough to scoop up a whale—supertrawlers are floating fish factories. These ships can catch and process more than 400 tons (360 t) of fish a day. No one knows how many fish swim the oceans. However, if too many are caught, some species may not recover. Is sustainable fishing possible with supertrawlers harvesting the seas?

Factories at Sea



State-of-the-art electronic gear allows super-trawlers to track schools of fish with pinpoint accuracy. Trawl nets may stretch half a mile behind the ships, engulfing everything in their paths.

The first factory trawler was built in Scotland in 1954. By 1970 the Soviet Union had 400 trawlers, the world's largest fleet at that time. Other countries, including the United States, China, and Japan, soon launched their own trawler fleets. Marine harvests soared as these enormous vessels went to work in the world's richest fishing grounds. But after a few years of bounty, there were signs of trouble at sea.

In the western Bering Sea, for example, Soviet (later Russian) supertrawlers initially harvested large numbers of sole, perch, herring, and especially pollock—the fish used in frozen fish sticks and fast-food fish sandwiches. As catches started to outpace reproduction rates, fish populations plummeted. Data gathered by Russian marine biologists show

A Russian supertrawler (below) searches for fish in the Sea of Okhotsk. Aboard these vessels, workers (right) prepare fish for a global market. ▼

that pollock catches are declining by 10 percent every year. American scientists raise similar concerns about the impact of American supertrawlers working the eastern Bering Sea.

Supertrawlers are usually after certain kinds of fish. Everything else hauled up in the nets gets discarded. Millions of fish and other marine animals die unnecessarily every year. Many trawlers also drag nets along the seafloor, destroying countless organisms and their habitats. Animals higher on the food chain are affected, too. Seals, sea lions, and kittiwakes can starve if there are fewer fish to eat. Since the 1970s these mammal and bird populations in the Bering Sea have declined.

Opponents of supertrawlers argue that the ships are doing irreparable damage to fish stocks and marine habitats. Opponents feel the unnecessary slaughter of healthy marine organisms is wasteful. Even though there are some restrictions on supertrawlers, opponents maintain that the



◀ Using a huge net, a Russian fisherman empties a load of fish into his boat.

laws are hard to enforce. Furthermore, since smaller boats can't compete with super-trawlers, the big ships threaten traditional fishing cultures on every continent.

Supporters of supertrawlers cite the growing global demand for fish and fish products. They point out that their catches supply high-protein food to millions of people. Some trawler operators dispute data that show a decline in fish populations. Others say that if trawlers reduce their catches, other ships will simply harvest what's left behind. Russian officials must balance the risk of destroying fish stocks with Russia's need for a profitable fishing industry.

What's Your Point of View?
Should further restrictions be placed on supertrawlers?
Should they be banned worldwide?



Guide to Reading

Consider What You Know

You have read about the Soviet government's development of heavy industry and about Russia's development of a market economy after the breakup of the Soviet Union. How do you think these activities have affected the quality of Russia's environment?

Read to Find Out

- How does Russia manage its natural resources?
- How has pollution affected the lives of Russia's people?
- What are the environmental challenges in Russia's future?

Terms to Know

- radioactive material
- pesticide
- nuclear waste

Places to Locate

- Kamchatka
- Lake Baikal

People and Their Environment



A Geographic View

The Aftermath

I found little likelihood that things [would] improve soon; the economies of Russia and most of the other . . . former Soviet republics are in shambles. "They used to show us films of the corrupted West with its polluted waters, like your Great Lakes," a Siberian environmental worker said. "Now the situation you had in the 1960s is here. But if the chaos continues, we will need two or three times as many years as you needed just to decide it's necessary to clean up."

—Mike Edwards, "Lethal Legacy,"
National Geographic, August 1994



Steel plant in Siberia

The world's expanding industries and rapidly growing population often strain the natural environment. Careless management of natural resources for short-term gain destroys economic opportunities for future generations, damages the environment, threatens people's health, and jeopardizes people's quality of life. In this section you will learn how Russia is managing its resources and balancing economic growth with environmental conservation.

Managing Resources

Russia is trying to make the best use of its vast and abundant natural resources in order to strengthen its economy and improve its standard of living. Unfortunately, the country has inherited a legacy of environmental damage. Russia's main challenge is to manage its resources without repeating its past disregard for the environment.

A second challenge is to improve the environment and repair damage that has already been done. One target for improvement is the timber industry. Russia contains the world's largest forest reserve, and the World Bank's Sustainable Forestry Pilot Project is helping Russia manage its forests more effectively. Using land more wisely, protecting forests, planting new trees, and increasing private forestry investment all help Russia's environment and economy. Higher taxes paid by Russian citizens provide income for the government to help protect the environment. Increased employment opportunities in the forest industry and more stable local economies will be possible only if steps to conserve the forest are taken. It is in the best interest of the people to protect the forests because the timber industry provides jobs and economic resources for communities.

Individual Russians are becoming more aware of the value of good environmental management. People have banded together to oppose a mining operation located in remote **Kamchatka** (kuhm•CHAWT•kuh), a region of Siberia in eastern Russia. The Kamchatka Committee for the Protection of the Environment and Natural Resources has demanded that the mining company meet strict environmental standards. The possible threat to the area's salmon spawning grounds prompted the local fishing industry to support the effort. The mine also caused concern among local residents and environmentalists because it was close to a nature park that was recently named a United Nations World Heritage site.

Pollution

The Soviets' disregard for the environmental effects of industrialization damaged Russia's water, air, and soil. By the 1990s, 40 percent of Russia's vast territory was under "ecological stress," with the health of millions of Russians affected by unchecked pollution and radiation.

Water Quality

Although Russia has one of the world's largest supplies of freshwater, industrialization has polluted most of its lakes and rivers. Fertilizer runoff, sewage, metals such as aluminum, and **radioactive material**—material contaminated by residue from the generation of nuclear energy—all contribute to

poor water quality. The waters of the Moskva and Volga Rivers, for instance, pose severe health risks. The many dams along the Volga have trapped contaminated water. Moscow has also reported cholera-causing bacteria in its water. Pollution even threatens the Caspian Sea.

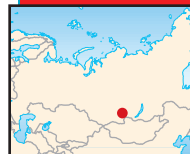
Lying on the southeastern edge of the Central Siberian Plateau, **Lake Baikal** (by•KAWL) is the world's oldest and deepest lake. It contains one-fifth of the world's freshwater, and 1,500 native species of aquatic plants and animals make their home there. Calling it "the Pearl of Siberia," Russians consider the lake a natural wonder. A recent traveler learned from a local resident what Baikal means to Russians:

"Lake Baikal is a symbol, Sasha told me once, of all the things that give Siberian life its distinct sweetness—the natural beauty, the purity of open air, the hardy generosity of people and the poetry in their collective soul. 'This is what Russians mean when they talk about the Motherland,' he said. 'And nothing, nothing is more precious to us than that.'"

Don Belt, "Russia's Lake Baikal: The World's Great Lake," *National Geographic*, June 1992

 NATIONAL GEOGRAPHIC

World Explorer



Geography

Skills for Life

Timber in Siberia Timber processing is a major economic activity along Siberia's Yenisey River.

Human-Environment Interaction How does the proper management of forests affect a country?



Geography Skills for Life

Lake Baikal Efforts to protect Lake Baikal include closing paper mills and installing wastewater treatment plants.

Human-Environment Interaction How has industrial development affected Lake Baikal?

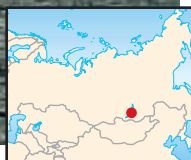
In 1957 the Soviet Union announced a plan to build a paper pulp factory in Baikalsk along Lake Baikal's southwestern shores. Although this plan was opposed by citizens in the area, their protests were ignored and the factory was built. This factory and others that followed continue to dump industrial waste into the lake. However, in response to the ongoing protests of local residents, the most serious polluters either have been closed or are in the process of reducing pollution. Pollution levels in the lake are now relatively low compared with many lakes in Europe and the United States.

Soil and Air Quality

For decades toxic waste dumps and airborne pollution poisoned Russia's soil. Aging storage containers cracked and leaked toxic wastes into the soil. Petroleum pipelines also often broke and tainted the land. Overuse of fertilizers and **pesticides**—chemicals used to kill crop-damaging insects, rodents, and other pests—has damaged farmland.

Russian experts believe that during the 1990s only 15 percent of Russia's urban population lived

with acceptable air quality. Industries, emissions from vehicles, and the soft coal burned for fuel are all sources of air pollution. In addition to releasing soot, sulfur, and carbon dioxide into the air, burning coal leads to another harmful agent—acid rain. Experts estimate that the combination of acid rain and chemical pollution has reduced Russian forests by about 1.5 million acres (607,500 ha) since the early 1970s.



Nuclear Wastes

Between 1949 and 1987, the Soviet Union set off more than 600 nuclear explosions. Soviets developed and then stockpiled nuclear weapons throughout the Cold War. Today, the condition and fate of those weapons concern Russia and the rest of the world.

Nuclear wastes are the by-products of producing nuclear power. Some of these wastes can remain radioactive for thousands of years, posing great dangers to people and the environment. The Soviets placed most nuclear wastes in storage facilities, but they also dumped some radioactive nuclear materials directly into Russia's northern waters, such as the Baltic and Bering Seas.

History

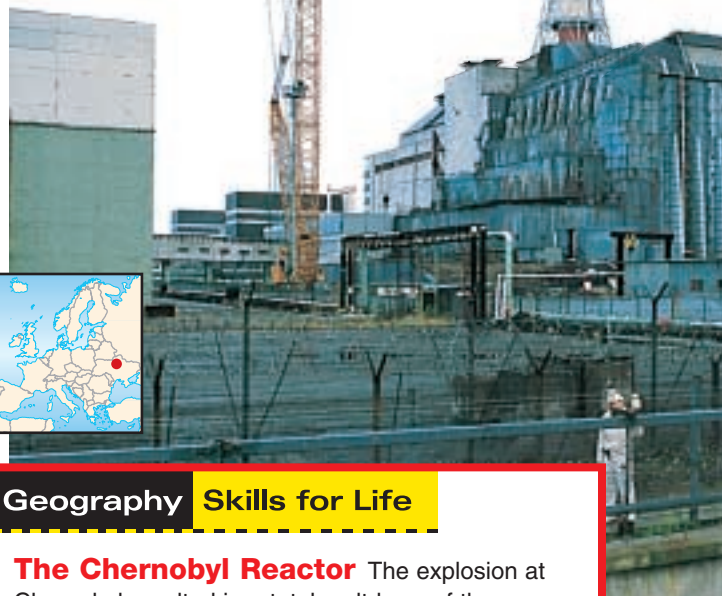
Chernobyl

During the Cold War, nuclear power generated much-needed electricity in the Soviet Union. It also provided power for building military weapons and vehicles. The urgency of keeping pace with the West during the Cold War often resulted in substandard nuclear plants and reactors that employed poorly trained workers who ignored proper safeguards. In 1986 a fire in a nuclear reactor in the town of Chernobyl (chuh•r•NOH•buhl), 60 miles (97 km) north of Kiev, Ukraine, released tons of radioactive particles into the local environment. This radiation was then carried great distances by the wind, and contaminated other countries.

Thousands of people were exposed to deadly levels of radiation because Soviet officials were slow to alert the public to the crisis and did not evacuate people soon enough. By the mid-1990s

over 8,000 people had died as a direct result of radiation poisoning. Millions more continue to suffer from cancer, stomach diseases, and immune system disorders. Radiation covered thousands of acres of farmland and forests in Belarus, Ukraine, and Russia. In Russia alone, radiation covered over 19,300 square miles (50,000 sq. km), where more than 30 million people lived. Because of prevailing winds, other countries suffered as well, most notably Finland, Sweden, Poland, and the former Czechoslovakia.

After the Chernobyl accident, international pressure prompted Soviet leaders to improve nuclear safety standards and to shut down dangerous plants. In response to these demands, Soviet officials never opened some newly built reactors and abandoned plans for building others. Despite concerns from other countries, 28 nuclear reactors continue to operate at nine sites throughout Russia. Much of Russia's electricity continues to come from these plants. In late 2000, however, the remaining reactor at Chernobyl was shut down. Experts in Western countries as well as in Russia and Ukraine think that many remaining Soviet-era reactors are poorly designed, unsafe, and should be made secure.



Geography

Skills for Life

The Chernobyl Reactor The explosion at Chernobyl resulted in a total meltdown of the core.

Human-Environment Interaction How did the Soviet government improve nuclear safety standards after the Chernobyl accident?

TAKS Practice

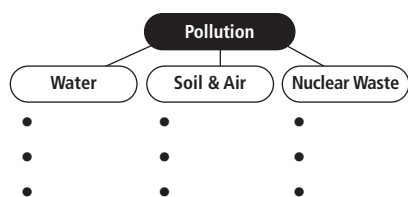
SECTION

2

ASSESSMENT

Checking for Understanding

- 1. Define** radioactive material, pesticide, nuclear waste.
- 2. Main Ideas** Create a graphic organizer like the one below, and fill in information about each of the topics. Then choose one of the topics, and summarize efforts currently under way in Russia to address the situation.



Critical Thinking

- 3. Making Generalizations** What generalizations can you make about the relationship between economic development and the environment in Russia?
- 4. Problem Solving** Assume the role of the Russian president, and identify an environmental problem in your country. What steps would you take to solve this problem?
- 5. Predicting Consequences** Think about what you know about the Russian economy. What is the likelihood of a dramatic improvement in Russia's environmental problems in the near future?

Analyzing Maps

- 6. Human-Environment Interaction** Study the economic activity map on page 341. Think about the regions of Russia in which pollution is a problem. Describe the relationship between the location of manufacturing centers and pollution.

Applying Geography

- 7. Influence of Location** Think about the challenges Russia faces concerning water quality. Write a paragraph explaining why Russians do not use more water from Lake Baikal to supply their freshwater needs.



**TAKS
Practice**

Categorizing Information

When you read a map, you make sense of the data you see—the symbols, words, and different-colored lines and shapes—by categorizing the information. Categorizing means grouping information and details together in a way that helps you understand and compare two or more ideas or concepts.

Learning the Skill

Categorizing information helps you make connections and retain information. This skill helps you answer questions such as *What is it? What parts does it have?* and *How is this like or unlike something else?*

When you categorize, you sort details into groups. You may be looking at a map, reading an informative article, or watching a basketball game. Once you understand how the details are grouped, you can make comparisons and draw conclusions. One way to keep track of the different details is to create a chart.

Follow these steps to categorize written information, using a chart.

- **As you read a section of a chapter, identify its main categories.** Make a two-column chart with one row for titles and one row for each category.
- **Spend a few minutes reading the section.** Record the title for each category in the first column of the chart. Then note some details and characteristics that you found for each category. List these in the second column of the chart.
- **Review the details in the second column of the chart.** Use them to write a summary statement about each category and to compare the categories with each other.

The Changing Economy of Russia

Type of Economy	Characteristics
Command (Soviet)	• Central authority owns banks, factories, farms, mines, and transportation systems. • Production and prices depend on decisions of the central authority. • Meets the basic needs of consumers but not designed to meet their wants.
Market	• Businesses are privately owned. • Production and prices depend on supply and demand. • A high degree of individual freedom allows producers to make whatever they think they will sell.

Practicing the Skill

Use the information about Russia on pages 390–392 and the chart on this page to answer the questions below.

1. What are the main categories of information on pages 390–392?
2. What are two other characteristics you could list in the chart?
3. How are these systems alike? How are they different?
4. What are two ways that a chart similar to the one above could help you?

Applying the Skill

Use library or Internet research and the information in Chapters 7 and 16 to categorize information about pollution in Russia, the United States, and Canada. Use a chart like the one on this page to list details about the sources of air, water, and soil pollution and proposed solutions for these challenges.



The **Glencoe Skillbuilder Interactive Workbook, Level 2** provides instruction and practice in key social studies skills.

CHAPTER 16

SUMMARY & STUDY GUIDE



SECTION 1

Living in Russia (pp. 387–393)

Terms to Know

- command economy
- consumer goods
- black market
- market economy
- privatization
- kolkhoz
- sovkhoz

Key Points

- The Soviet economy was a command economy controlled by government agencies.
- Since the 1980s the Russians have been making the difficult transition from the Soviet command economy to a market economy.
- After the breakup of the Soviet Union, Boris Yeltsin encouraged privatization of state-owned farms and businesses.
- Transportation and communications systems must improve in order to support a strong market economy.
- To take its place as a full partner in the global community, Russia needs good international trade and strong political and economic relations.

Organizing Your Notes

Create an outline like the one below, using the section headings to help you organize your notes for this section.

Living in Russia	
I. Changing Economies	
A. The Soviet Command Economy	
1.	
2.	

SECTION 2

People and Their Environment (pp. 396–399)

Terms to Know

- radioactive material
- pesticide
- nuclear waste

Key Points

- Soviet leaders' drive for an industrial-based economy caused major and lasting damage to Russia's water, soil, and air.
- Russia needs to manage its use of natural resources properly in order to avoid more environmental damage.
- Radioactivity from nuclear waste, nuclear accidents, and aging nuclear weapons poses a grave danger to Russia's environment and its people's health.

Organizing Your Notes

Use a graphic organizer like the one below to help you organize information about the challenges facing Russia today.

Environmental Challenges		
Water	Soil and Air	Nuclear Waste

Russian passengers wait to board a train.



CHAPTER 16

ASSESSMENT & ACTIVITIES

Reviewing Key Terms

On a sheet of paper, classify each of the lettered terms below into the following categories. (Some terms may apply to both categories.)

- | | |
|--------------------|-------------------------|
| • Soviet Era | • After Independence |
| a. command economy | f. kolkhoz |
| b. consumer goods | g. sovkhov |
| c. black market | h. radioactive material |
| d. market economy | i. pesticide |
| e. privatization | j. nuclear waste |

Reviewing Facts

SECTION 1

1. What was the role of the government in the Soviet command economy?
2. How has the transition to a market economy challenged Russian society?
3. How did privatization impact daily life in Russia?
4. What steps has Russia taken to become part of the global community?

SECTION 2

5. What problems have been created by pollution in Russia?
6. What challenges with the environment and natural resources does Russia face today?
7. How did the Cold War contribute to Russia's environmental problems?

Critical Thinking

1. **Making Inferences** Study the chart on pages 388–389. How might political and economic reforms in Russia eventually affect the distribution of the country's resources?

2. **Problem Solving** Identify one kind of pollution affecting Russia. Describe the cause or origin of the pollution. What steps do you think would be necessary to reduce its effects? Explain the reasons for your answer.

3. **Categorizing Information** Complete a diagram like the one below to show the changes in Russian life after independence. Then write a paragraph explaining the change you think had the greatest impact on Russian life.

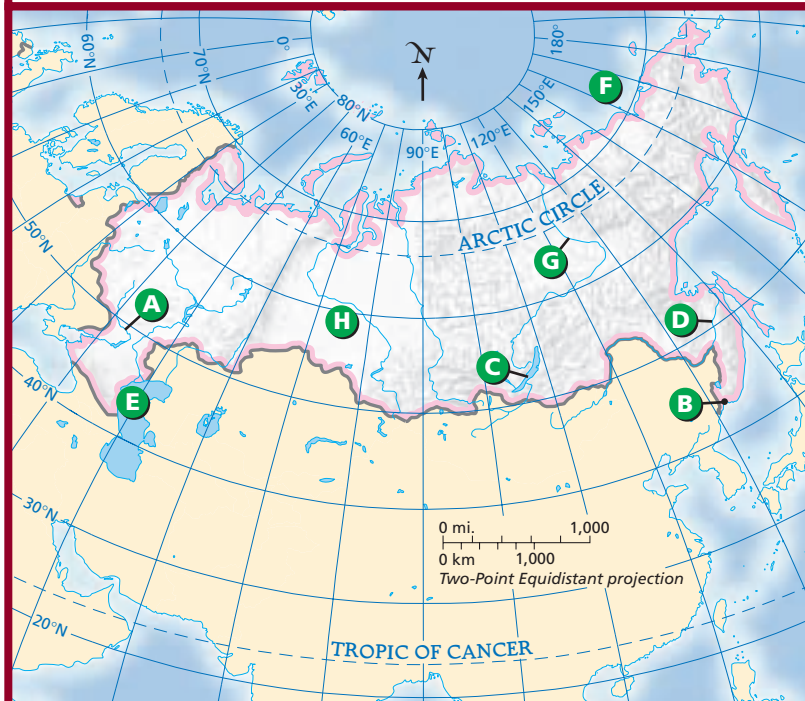


Locating Places

Russia: Physical-Political Geography

Match the letters on the map with these places and physical features of Russia. Write your answers on a sheet of paper.

- | | | |
|----------------------|------------------------|----------------|
| 1. Don River | 4. West Siberian Plain | 6. Amur River |
| 2. Caspian Sea | 5. Lena River | 7. Vladivostok |
| 3. East Siberian Sea | | 8. Lake Baikal |



Self-Check Quiz Visit the **Glencoe World Geography** Web site at tx.geography.glencoe.com and click on Self-Check Quizzes—Chapter 16 to prepare for the Chapter Test.

CLICK HERE

Using the Regional Atlas

Refer to the Regional Atlas on pages 338–341.

- 1. Location** Which manufacturing areas are located along the Volga River? Along the Ob and Irtysh Rivers?
- 2. Human-Environment Interaction** Compare the population density map and the economic activity map. Describe the correlation between commercial farmland and population density.

Thinking Like a Geographer

Think about the population distribution in Russia. How do you think the shift from a command economy to a market economy might affect migration patterns and population density? Write a paragraph about the most significant effect.

Problem-Solving Activity

Contemporary Issues Case Study Russia remains an influential international power despite its recent political and economic changes. Learn more about Russia's policies toward the expansion of the European Union. Then focus on ways Russian and European interests are similar and different. In an essay, present your conclusions about the future of Russian-European relations.

GeoJournal

Expository Writing Use your GeoJournal, your textbook, and the Internet to research and write an essay that analyzes how Moscow's character as a city is related to its political, social, economic, and cultural features.



Technology Activity

Developing Multimedia Presentations Compile information about people's work, school, community, home life, and leisure activities in Russia. Use various media to introduce Russian life to other students or people in your community. For example, you might play recordings of popular Russian music or show a film that presents an aspect of Russian culture, in addition to your oral report.



TX TAKS Test Practice

Use your knowledge of Russia to choose the best answer for each of the following multiple-choice questions. If you have trouble answering the questions, use the process of elimination to narrow your choices.

- 1. In Russia which of these challenges affects the transportation of both petroleum products and other goods?**

- A Poorly repaired roads
- B Harsh weather and vast distances
- C Frozen waterways
- D Separatist movements



First determine what choices you can eliminate. Since petroleum products are transported through pipelines, choices A and C do not apply and can be eliminated. Choose the best answer from the remaining options.

- 2. In Russia, nuclear power plants built during the Soviet era**

- F have been shut down.
- G provide much of Russia's electricity.
- H are now safer than ever before.
- J have been replaced by coal-fired generators.



Only one answer is completely true. Some reactors have been shut down, and some safety standards were improved. Choose the answer that is completely true.